

PROFESSIONAL SUMMARY

Full Stack Software Engineer with 3+ years of experience building scalable, enterprise-grade systems across public-sector and product-based environments. Skilled in developing RESTful APIs, architecting microservices, and delivering cloud-native applications using Java, Spring Boot, Angular, SQL, and AWS. Specialise in improving system performance, code quality, and delivery speed within Agile teams, with a strong track record of leading development initiatives and optimising complex system workflows.

TECHNICAL SKILLS

Languages | Java, Python, SQL, JavaScript, TypeScript, NoSQL
Backend | Spring Boot, Hibernate, Flask, RESTful APIs, Microservices Architecture
Frontend | Angular, React, Bootstrap, HTML, CSS, Vue
Cloud & DevOps | AWS (EC2, S3, RDS, Lambda, Amplify), Jenkins, Docker, Kubernetes, CI/CD Pipelines
Testing | JUnit, JaCoCo, Test-Driven Development (TDD), Integration & API Testing
Design & Architecture | SOLID Principles, Design Patterns, Scalable System Design
Tools | Jira, Git, Postman, Swagger, Vercel, Netlify
Operating Systems | Linux, Windows, MacOS

WORK EXPERIENCE

Development Lead

Aug 2024 - Present

Jira | Git

NCS Group, Singapore | National Harmonised Integrated Pharmacy System (NHIPS)

- Own end-to-end User Story delivery, driving sprint planning, backlog refinement, and task breakdowns to consistently achieve **100%** sprint completion
- Strengthen requirement clarity by facilitating detailed grooming discussions, mitigating project ambiguity by **15%** and minimising downstream rework
- Mentor a team of **3** developers by assigning tasks, removing blockers, and tracking progress, achieving **95%** delivery consistency over **12** months
- Act as a key liaison in sprint demos, ensuring alignment between Product Owners and engineering teams, improving PO satisfaction by **20%**

Software Engineer (Backend)

Aug 2022 - Aug 2024

Java | SpringBoot | Hibernate | SQL | JUnit

NCS Group, Singapore | National Harmonised Integrated Pharmacy System (NHIPS)

- Built and optimised backend services for a national pharmacy system, delivering robust RESTful APIs and scalable Java modules supporting **1,000+** daily users
- Drove integration of NHIPS with **50+** pharmacy machines using API connectors and middleware components, enhancing operational interoperability
- Improved query performance by implementing indexed foreign keys and aggregated SQL functions, streamlining medication order creation time by **37%**
- Implemented automated unit testing with JUnit and JaCoCo, ensuring **90%+** coverage on new features and maintaining project-wide coverage above **80%**

Software Engineer (Frontend)

May 2021 - Jan 2022

Typescript | Angular | Bootstrap | CSS

NCS Group, Singapore | National Harmonised Integrated Pharmacy System (NHIPS)

- Developed modular, reusable Angular components to standardise UI behaviours across **12+** modules, standardising development and improving maintainability
- Engineered frontend performance strategies including lazy loading and smart caching, achieving a **25%** decrease in page load times
- Contributed to major refactoring efforts that eliminated redundant logic and reduced technical debt by **20%**, enhancing stability and scalability
- Enhanced data-heavy UIs by leveraging advanced Ag-Grid configurations, optimising rendering speeds by **40%** for high-volume datasets

Machine Learning Intern

May 2021 - Jan 2022

Python | MSSQL | BERT

BeLive Technology, Singapore | BeLive Audience Sentiment Engine (B.A.S.E)

- Assisted in the development of the BeLive Audience Sentiment Engine (B.A.S.E) to process and provide useful insights of a host/business live stream
- Trained and evaluated BERT-based sentiment models for livestream analytics, achieving **93.9%** accuracy, delivering actionable insights to business users
- Improved dataset quality through strategic relabelling, optimising accuracy by over **3%** across multiple model versions
- Collaborated with cross-functional teams to extend the Engine's capabilities to multiple languages, leading to a **27%** boost in international user engagement

Data Science Intern

Aug - Nov 2021

Python | SKLearn | TF-IDF

BeLive Technology, Singapore | Question & Language Detection Models

- Employed Regular Expression (Regex) techniques to detect questions within livestream comments, attaining a robust testing accuracy of **>91%**
- Utilised Vectorisation to accurately identify the language in livestream comments across **3** languages: English, Chinese, and Bahasa
- Launched automated sentiment analysis on livestream data, empowering hosts with real-time insights, growing audience engagement by **23%**
- Strategically relabeled livestream comment datasets to fine-tune models, delivering an overall accuracy of **>96%**, providing more precise insights to businesses

Project Manager

2021 - 2022

React | Python | Flask | Couchbase

CitiVentures Innovation Lab, Singapore | Predictive Analysis of Risk Utilisation (Phase II)

- Initiated the end-to-end development of the Application UI, orchestrating a **15%** reduction in UI development timeline, ensuring timely system delivery
- Led the creation of an interactive dashboard tailored to Citibank Risk Managers, accelerating response time to address compliance concerns by **40%**
- Designed **3** distinct Machine Learning solutions harnessing historical trading patterns, achieving a **23%** rise in risk parameter prediction accuracy
- Initiated workshops promoting cross-departmental understanding of the developed solution, facilitating a **27%** increase in interdepartmental collaboration

Other Projects

2021 - 2022

- Project Peter:** Used pre-trained BERT Model architectures and Natural Language Processing to create a Question-Answering Chatbot that can answer questions
- Auto Artist:** Created and trained a Denoising Autoencoder Model using TensorFlow and Keras to detect and patch damaged images that have cracks
- Seah Kim Cheok:** Performed advanced rule-based algorithms to analyse incoming, real-time vehicular data, identifying accidents with an accuracy rate of **92%**
- Open World Tourism:** Converted GLTF models into declarative, reusable JSX components to create significant landmarks (e.g., Merlion) within the application

EDUCATION

National University of Singapore

2022 - Present

Master of Technology in Software Engineering (MTech) - Architecting Scalable Systems, Microservices, Cloud-native Development

Certifications: Graduate Certificate in Architecting Scalable Systems, Graduate Certificate in Architecting AI Systems

Singapore Management University

May 2022

Bachelor of Science (Information Systems) - Financial Technology & Artificial Intelligence

Honours: 2 x Dean's List Recipient, Teaching Assistant for modules in Software Project Management, Social Analytics & Management Communications

CERTIFICATIONS

Professional Scrum Master (PSM I) , Amazon Web Services Certified Cloud Practitioner (CLF-C01) , Google AI Explore ML Beginner Track